Supplementary material

Mutants overstate what LLM-written unit tests can find: reach on mutants and on real faults · Ook Lee

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S1 Per-module mutation results

Table S1 Mutants and kill rates per module. Kill rate = share of all mutants killed. Modules with fewer than 30 mutants are shown but excluded from the reach estimates in the main text.

Project	Module	Mutants	Existing tests kill	Haiku K=1 / K=8	Sonnet K=1 / K=8	DeepSeek K=1 / K=8	In main analysis
boltons	boltons/ iterutils.py	150	0.55	0.69 / 0.80	0.71 / 0.75	0.68 / 0.77	yes
boltons	boltons/ listutils.py	120	0.66	0.68 / 0.90	0.69 / 0.82	0.44 / 0.85	yes
boltons	boltons/ strutils.py	150	0.47	0.65 / 1.00	0.65 / 0.83	0.57 / 0.77	yes
cachetools	src/cachetools/ func.py	21	0.76	0.81 / 0.95	0.48 / 1.00	0.48 / 0.95	no (< 30 mutants)
cachetools	src/ cachetools/ keys.py	5	1.00	1.00 / 1.00	1.00 / 1.00	1.00 / 1.00	no (< 30 mutants)
humanize	src/humanize/ filesize.py	20	0.95	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	no (< 30 mutants)
humanize	src/humanize/ number.py	144	0.90	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	yes
more-itertools	more_itertools/ recipes.py	150	0.82	0.63 / 0.69	0.68 / 0.71	0.53 / 0.65	yes
toolz	toolz/ dicttoolz.py	18	0.89	0.89 / 0.89	0.83 / 0.83	0.83 / 0.83	no (< 30 mutants)
toolz	toolz/ functoolz.py	123	0.72	0.64 / 0.66	0.62 / 0.72	0.46 / 0.63	yes
toolz	toolz/ itertoolz.py	110	0.93	0.87 / 0.95	0.94 / 0.95	0.69 / 0.95	yes

Table S2 Mutants killed only by the LLM tests (K = 8), as a share of all mutants of the module.

Module	Haiku: LLM-only share	Sonnet: LLM-only share	DeepSeek: LLM-only share
boltons/iterutils.py	0.29	0.26	0.28
boltons/listutils.py	0.24	0.21	0.21
boltons/strutils.py	0.53	0.35	0.33
more_itertools/recipes.py	0.02	0.03	0.02
src/humanize/number.py	0.00	0.00	0.00
toolz/functoolz.py	0.07	0.08	0.05
toolz/itertoolz.py	0.05	0.05	0.05

S2 Reach on mutants by number of attempts

Table S3 Reach on mutants relative to the existing tests (module-level ratio among mutants killed by either side, averaged over 7 modules; 95% intervals from 1,500 module resamples).

K	Claude Haiku 5.5	Claude Sonnet 5.5	DeepSeek V4.1 Flash

1	0.90 (0.57–1.16)	0.92 (0.58–1.19)	0.74 (0.46–1.01)
2	1.05 (0.60–1.49)	0.98 (0.62–1.29)	0.86 (0.53–1.16)
3	1.07 (0.62–1.51)	1.00 (0.61–1.30)	0.91 (0.57–1.22)
4	1.09 (0.65–1.53)	1.02 (0.65–1.34)	0.94 (0.57–1.24)
5	1.09 (0.67–1.54)	1.03 (0.64–1.35)	0.97 (0.58–1.30)
6	1.09 (0.68–1.51)	1.03 (0.61–1.36)	0.97 (0.61–1.31)
7	1.10 (0.67–1.53)	1.04 (0.63–1.36)	1.00 (0.64–1.33)
8	1.10 (0.64–1.56)	1.04 (0.68–1.36)	1.01 (0.58–1.34)

S3 Per-project real-fault results

Table S4 Share of real faults revealed, by project.

Project	Faults	Haiku K=1 / K=6	Sonnet K=1 / K=6	DeepSeek K=1 / K=6	Any model K=6
youtube-dl	38	0.05 / 0.08	0.05 / 0.08	0.05 / 0.13	0.16
thefuck	32	0.03 / 0.19	0.09 / 0.12	0.09 / 0.16	0.31
tqdm	7	0.14 / 0.29	0.14 / 0.29	0.14 / 0.29	0.43
httpie	5	0.00 / 0.20	0.00 / 0.00	0.00 / 0.20	0.20
fastapi	3	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00
tornado	1	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00
cookiecutter	1	0.00 / 0.00	0.00 / 0.00	0.00 / 0.00	0.00
All	87	0.05 / 0.14	0.07 / 0.10	0.07 / 0.15	0.23

Table S5 Validation of BugsInPy bugs. A bug is valid if the developer's fault-revealing test fails on the faulty and passes on the fixed version in our environment. Invalid bugs mostly failed to install with their pinned dependencies or did not reproduce the failure.

Project	Bugs tried	Valid (fails on faulty, passes on fixed)
youtube-dl	43	38
thefuck	32	32
tqdm	9	7
httpie	5	5
fastapi	16	3
cookiecutter	4	1
tornado	16	1
PySnooper	3	0
sanic	5	0
luigi	33	0
black	23	0
scrapy	40	0

S4 Per-bug results

Table S6 Number of the six attempts that revealed the fault, per bug and model.

Bug	Haiku: attempts revealing (of 6)	Sonnet: attempts revealing (of 6)	DeepSeek: attempts revealing (of 6)
cookiecutter/4	0	0	0
fastapi/1	0	0	0
fastapi/2	0	0	0
fastapi/3	0	0	0
httpie/1	0	0	0

httpie/2	0	0	0
httpie/3	0	0	0
httpie/4	0	0	0
httpie/5	2	0	1
thefuck/1	0	0	0
thefuck/2	0	0	0
thefuck/3	0	0	0
thefuck/4	0	0	0
thefuck/5	0	0	0
thefuck/6	0	0	0
thefuck/7	1	0	0
thefuck/8	1	0	0
thefuck/9	1	0	0
thefuck/10	0	0	0
thefuck/11	0	0	0
thefuck/12	0	0	0
thefuck/13	2	0	1
thefuck/14	0	0	1
thefuck/15	0	0	0
thefuck/16	0	0	0
thefuck/17	0	0	0
thefuck/18	0	0	0
thefuck/19	0	0	0
thefuck/20	0	0	0
thefuck/21	0	0	1
thefuck/22	2	6	2
thefuck/23	0	0	0
thefuck/24	0	1	0
thefuck/25	0	0	0
thefuck/26	0	0	0
thefuck/27	0	0	0
thefuck/28	2	3	1
thefuck/29	0	0	0
thefuck/30	0	1	0
thefuck/31	0	0	0
thefuck/32	0	0	0
tornado/9	0	0	0
tqdm/2	4	6	6
tqdm/3	0	0	0
tqdm/5	1	0	2
tqdm/6	0	0	0
tqdm/7	0	0	0
tqdm/8	0	2	0
tqdm/9	0	0	0
youtube-dl/1	0	0	0
youtube-dl/2	0	0	0
youtube-dl/3	0	0	0
youtube-dl/4	0	0	0
youtube-dl/5	0	1	2
youtube-dl/6	0	0	0
youtube-dl/7	0	0	0
youtube-dl/8	0	0	0
youtube-dl/9	0	0	0
youtube-dl/10	0	0	0
youtube-dl/11	0	0	0
youtube-dl/12	0	0	0
youtube-dl/13	0	0	0
youtube-dl/14	0	0	0
youtube-dl/15	0	0	0
youtube-dl/16	0	0	0
youtube-dl/17	0	0	0
youtube-dl/18	0	0	0
youtube-dl/19	0	0	0
youtube-dl/20	0	0	0

youtube-dl/21	0	0	0
youtube-dl/22	0	0	0
youtube-dl/23	0	0	0
youtube-dl/24	0	0	0
youtube-dl/25	0	0	0
youtube-dl/26	0	0	0
youtube-dl/27	0	0	0
youtube-dl/28	0	0	0
youtube-dl/29	0	0	2
youtube-dl/30	0	0	0
youtube-dl/31	1	0	0
youtube-dl/32	0	0	0
youtube-dl/33	0	0	2
youtube-dl/34	0	0	0
youtube-dl/35	0	0	0
youtube-dl/37	0	0	0
youtube-dl/41	5	6	4
youtube-dl/43	1	1	1

S5 Prompts

Mutation study (one call per attempt; `{modname}` is the module's import path, followed by the module source in a python code block):
Below is the source of the Python module `{modname}`. Write a thorough pytest test file for it. Test the behaviour of its public functions, including edge cases. Import from `{modname}`. Keep it under 250 lines. Output only one python code block.
Real-fault study (one call per attempt; followed by the non-test files changed by the fix, in their faulty state, each prefixed by '# file: <path>' and truncated to 40,000 characters):
Below are source files from the Python project `{project}`. Write a thorough pytest test file for them. Test the behaviour of the public functions and classes, including edge cases. Use only absolute imports of the project's modules. Keep it under 250 lines. Output only one python code block.
Code was extracted from the first python code block of the answer. Models were called at their default temperature with up to 16,000 output tokens.

S6 Procedures, environment and cost

Mutants. Operators: arithmetic operator replacement, comparison operator replacement, Boolean operator replacement, constant replacement and negation removal on the module's AST; at most 150 mutants per module, sampled with a fixed seed. Generated tests that failed on the original module were discarded; a mutant is killed if a retained test fails or errors on it (with a timeout).
Real faults. Each bug was checked out from a bare clone of the project at the faulty commit; the fixed version was obtained by applying the BugsInPy patch. Each project ran in its own virtual environment with the Python version recorded by BugsInPy (3.8 where older versions were unavailable) and setuptools < 60. A generated test reveals the fault if it fails on the faulty and passes on the fixed version; failures caused by import or collection errors are not counted. Test files that hung were stopped by a timeout and counted as producing no usable test.
Statistics. Mutation reach: module-level ratios averaged across modules; 95% percentile intervals from 1,500 resamples of modules. Real-fault reach: share of faults revealed; 95% percentile intervals from 3,000 two-stage resamples (projects, then bugs within projects).
Cost. API spending was about US$28 for the real-fault study (87 bugs × 3 models × 6 attempts) and is recorded with the mutation study's ledger in the replication package.